

# Build Your Research Portfolio Website From Scratch

## A Complete Step-by-Step Guide Using Quarto + GitHub Pages

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# 1 Understanding the Basics

## 1.1 What Are We Building?

Before touching the keyboard, let's understand what we're doing in plain language. Your portfolio website is just a collection of text files on your computer. **Quarto** reads those files and converts them into a beautiful website. **GitHub** is a website where you store your files online — and it automatically shows them as a website to anyone in the world.

**Key idea:** You write content in simple text files → Quarto turns them into a website → GitHub shows that website to the world — for free.

## 1.2 The Four Tools You Will Use

You will install and use exactly four tools:

| Tool           | What It Does                                                |
|----------------|-------------------------------------------------------------|
| <b>Git</b>     | Saves and tracks your files; sends them to GitHub           |
| <b>VS Code</b> | A free text editor — you write all your content here        |
| <b>Quarto</b>  | Converts your text files into a beautiful website           |
| <b>GitHub</b>  | Free online service that hosts your website on the internet |

## 1.3 What Is a Terminal / Command Prompt?

A terminal (also called command prompt) is a text-based way to control your computer. Instead of clicking buttons, you type commands. Don't be scared — you will only use about 6 simple commands in this whole guide.

**How to open it on Windows:** Press the Windows key → type `cmd` → press Enter. A black window opens. That's your terminal. Type commands there and press Enter to run them.

**Important:** Never close the terminal while a command is still running. Wait for the cursor (>) to appear again before typing the next command.

## 1.4 What Your Finished Website Will Look Like

Your portfolio will have these pages:

- **Home** — Your photo, name, bio, and research interests
- **About** — Detailed background, education, and experience
- **Research** — All your papers with links and personal notes
- **Blog** — Short research notes and field observations

Your website address will be: <https://yourusername.github.io> — free, forever.

---

## 2 Installing the Tools (Windows)

You need to install three tools. Do them in order — do not skip ahead.

### 2.1 Install Git

Git is a tool that tracks every change you make to your files — like a time machine. It also lets you send your files to GitHub.

**Step 1** — Open your browser and go to:

```
https://git-scm.com/download/win
```

**Step 2** — The download will start automatically. If it doesn't, click the link that says "Click here to download manually".

**Step 3** — Run the installer. Click **Yes** if Windows asks permission. Click **Next** → **Next** → **Next** → **Install** → **Finish**. Do NOT change any options. All the default settings are correct.

**Step 4** — Verify it worked. Open a **NEW** terminal window and type:

```
git --version
```

You should see something like:

```
git version 2.44.0.windows.1
```

**If you see "command not found":** Restart your computer and try again.

### 2.2 Install VS Code

VS Code is a free text editor made by Microsoft. You will use it to write and edit all your website files.

**Step 1** — Go to:

```
https://code.visualstudio.com
```

**Step 2** — Click the big blue **Download for Windows** button. Run the downloaded installer.

**Step 3** — Go through the installer. On the screen that says “**Select Additional Tasks**” — tick these two boxes:

- ☒ Add “Open with Code” action to Windows Explorer file context menu
- ☒ Add to PATH

Then click **Next** → **Install** → **Finish**.

**Step 4** — VS Code will open automatically. Now install the Quarto extension:

- Look at the left side of VS Code — click the **Extensions** icon (4 small squares)
- In the search box, type: **Quarto**
- Click the first result (by quarto.org) → click **Install**

**Tip:** The Quarto extension gives you syntax highlighting, live preview, and helpful autocomplete when writing your website files.

## 2.3 Install Quarto

Quarto is the tool that reads your text files and builds your actual website.

**Step 1** — Go to:

```
https://quarto.org/docs/get-started
```

**Step 2** — Click the download button for Windows. Run the file that downloads (ends in **.msi**).

**Step 3** — Click through the installer with all default options.

**Step 4** — **IMPORTANT:** Close your terminal and open a **NEW** one. Old terminal windows will not recognise newly installed tools.

**Step 5** — Verify it worked:

```
quarto --version
```

You should see something like: **1.4.557**

---

## 3 Setting Up GitHub

GitHub is the free service that will host your website. You need one account and one “repository” (a folder on GitHub where your files live).

### 3.1 Create Your GitHub Account

**Step 1** — Go to:

```
https://github.com
```

**Step 2** — Click **Sign up**. Fill in:

- Your email address
- A password
- A **username** — choose this carefully, it becomes part of your website address

**Username tips:** Use your real name or a professional variation. No spaces allowed — use a dash or nothing. Example: `subhradip25` or `subhradip-bhattacharjee` Your website will be: <https://subhradip25.github.io>

**Step 3** — GitHub will send a verification code to your email. Check your inbox and enter the code to activate your account.

### 3.2 Create Your Repository

A repository is like a project folder on GitHub.

**Step 1** — After signing in, click the + icon at the top right → **New repository**

**Step 2** — In the “Repository name” box, type **EXACTLY**:

```
yourusername.github.io
```

Replace `yourusername` with your actual GitHub username. For example: `subhradip25.github.io`

**Step 3** — Make sure **Public** is selected (not Private).

**Step 4** — Scroll down and click the green **Create repository** button.

**Why this special name?** The name `yourusername.github.io` tells GitHub to automatically serve this repository as a live website at <https://yourusername.github.io> — completely free!

---

## 4 Creating Your Quarto Project

Now we create the actual website project on your computer.

### 4.1 Create the Project

**Step 1** — Open VS Code. Click **Terminal** in the top menu → **New Terminal**. A panel opens at the bottom. This is your terminal inside VS Code.

**Step 2** — Navigate to your Documents folder:

```
cd Documents
```

**Step 3** — Create a new Quarto website project:

```
quarto create project website myportfolio
```

Quarto will ask one question about the directory name. Just press **Enter** to accept.

**Step 4** — Move into the project folder:

```
cd myportfolio
```

**Step 5** — Open the project in VS Code:

```
code .
```

**What does `code .` mean?** The dot ( `.` ) means “open the current folder”. VS Code will open with all your project files listed in the left panel.

## 4.2 Understanding Your File Structure

After creating the project, you will see these files on the left side of VS Code:

| File                     | Purpose                                             |
|--------------------------|-----------------------------------------------------|
| <code>_quarto.yml</code> | Master settings — menu, title, theme, output folder |
| <code>index.qmd</code>   | Your home page — bio, photo, research interests     |
| <code>about.qmd</code>   | Your detailed about/CV page                         |
| <code>styles.css</code>  | Custom colours and design tweaks                    |

## 4.3 Preview Your Site Locally

Before making any changes, let’s see what your site looks like right now.

**Step 1** — In the VS Code terminal, type:

```
quarto preview
```

**Step 2** — Wait 10–15 seconds. Your browser will automatically open and show your website. This is your **local preview** — only visible on your computer.

**Step 3** — When you want to stop the preview, press **Ctrl+C** in the terminal.

**Live reload:** While the preview is running, every time you save a file (Ctrl+S), the browser automatically refreshes to show your changes. You don’t need to restart the preview each time.

---

# 5 Building Your Home Page

The home page is the first thing visitors see — your name, photo, bio, and research interests.

## 5.1 Edit `_quarto.yml` — Site Settings

This file controls your site’s title, menu structure, and visual theme. Click on `_quarto.yml` in the left panel, **select all text** (Ctrl+A), **delete it**, and paste this:

```
project:
  type: website
```

```

output-dir: docs

website:
  title: "Your Full Name"
  navbar:
    left:
      - href: index.qmd
        text: Home
      - href: about.qmd
        text: About
      - href: research.qmd
        text: Research
      - href: blog.qmd
        text: Blog

  format:
    html:
      theme: cosmo
      css: styles.css
      toc: true

```

Press **Ctrl+S** to save. Your browser preview will update automatically.

**Tip:** You can change the `theme` from `cosmo` to other options like `flatly`, `litera`, `sandstone`, or `journal` to change the look of your site.

## 5.2 Edit `index.qmd` — Home Page

Click on `index.qmd`, **select all**, **delete**, and paste this template. Then fill in your own details:

```

---
title: "Your Full Name"
subtitle: "Geospatial & Agronomy Researcher"
image: images/profile.png
about:
  template: jolla
  links:
    - icon: envelope
      text: Email
      href: mailto:your@email.com
    - icon: mortarboard-fill
      text: Google Scholar
      href: https://scholar.google.com/your-profile
    - icon: file-person
      text: ResearchGate
      href: https://researchgate.net/your-profile

```

```

- icon: linkedin
  text: LinkedIn
  href: https://linkedin.com/in/yourprofile
---

I am a researcher at **Your Institution**, specialising in
geospatial analysis and agronomy.

My work focuses on understanding agricultural systems using
remote sensing, GIS, and field-based approaches.

## Research Interests

- Geospatial analysis of agricultural landscapes
- Coastal agronomy and soil health
- Remote sensing for crop monitoring
- Climate impacts on farming systems

```

Press **Ctrl+S** to save and check your browser.

### 5.3 Add Your Profile Photo

**Step 1** — Create an `images` folder in the VS Code terminal:

```
mkdir images
```

**Step 2** — Open Windows File Explorer. Navigate to:

```
Documents → myportfolio → images
```

Copy your photo into this folder and **rename it** to exactly: `profile.png` (or `profile.jpg` if your photo is a JPG file)

**Step 3** — In `index.qmd`, make sure this line matches your file extension:

```
image: images/profile.png # for PNG files
image: images/profile.jpg # for JPG files
```

**Step 4** — Save and check your browser. Your photo will appear in a circle.

**Photo tips:** Use a clear, well-lit, professional photo. Portrait or square orientation works best.

---

## 6 Building Your Research Page

The research page is the most important page for a researcher. It lists all your publications with links, co-authors, and your personal plain-language notes.

## 6.1 Create research.qmd

In VS Code, click **File** → **New File** → name it `research.qmd` → save it inside your `myportfolio` folder.

## 6.2 Format for Each Paper

Use this template for each paper. Copy and repeat it for every publication:

```
---
title: "Research"
---

## Journal Articles

---

**Your Paper Title Here** (2024)
*Journal Name* | Co-authors: Name A, Name B
[Read Paper](https://doi.org/your-doi-link) | [ResearchGate](https://researchgate.net)

> In simple words: Write 2–3 lines here explaining what this paper
> is about and why it matters. This is your personal note for visitors
> – not the abstract.

---
```

## 6.3 Export All Papers from Google Scholar

If you have many papers, don't type them manually. Use this shortcut:

**Step 1** — Go to your Google Scholar profile and sign in.

**Step 2** — Tick the **checkbox** at the top to select all papers. A blue bar will appear.

**Step 3** — Click **Export** → select **BibTeX** → a `.bib` file downloads with all your paper details.

**Step 4** — Open `claude.ai`, paste the BibTeX content, and ask:

*“Format all these papers into a `research.qmd` file for a Quarto portfolio”*

Claude will generate the complete, properly formatted page instantly.

**Suggested structure:** Organise by type: Journal Articles → Book Chapters → Conference Papers → PhD Thesis. Put newest papers first within each section.

---

## 7 Building Your Blog

The blog is where you share short research notes, field observations, and methodology explainers — content that is valuable but not suited for a formal paper.

## 7.1 Create the Blog Structure

**Step 1** — Create `blog.qmd` in your `myportfolio` folder with this content:

```
---
title: "Blog"
listing:
  contents: posts
  sort: "date desc"
  type: default
  categories: true
---
```

**Step 2** — Create the `posts` folder in the terminal:

```
mkdir posts
```

**Step 3** — Create a folder for your first post:

```
mkdir posts\first-post
```

**Step 4** — In VS Code, right-click the `first-post` folder → **New File** → name it `index.qmd`

## 7.2 Write Your First Blog Post

Paste this template into `posts/first-post/index.qmd`:

```
---
title: "Your Post Title Here"
date: "2025"-05-01
categories: [remote sensing, agronomy]
---
```

Write your post content here in plain English.

You can use:

- Bullet points
- **\*\*Bold text\*\***
- *\*Italic text\**

Add an image like this:

**![Caption text](../.. /images/your-image.png)**

**Blog ideas for researchers:** Field observations • Methodology notes • Literature review summaries  
• Conference takeaways • Software tutorials • Data analysis tips • Explaining your paper in simple language

## 7.3 Adding More Posts Later

For every new blog post, follow the same pattern:

```
# Create a folder for the new post
mkdir posts\your-post-name

# Then in VS Code, create: posts/your-post-name/index.qmd
# Fill in title, date, categories, and content
```

**Auto-sorted:** Quarto automatically sorts posts by date (newest first) and creates a category filter. You don't need to do anything extra.

---

## 8 Publishing Your Site Live

This is the exciting part — making your site publicly visible on the internet. This is a one-time setup. After this, updating takes just 4 commands.

### 8.1 Step 1 — Tell Git Who You Are

Run these two commands in the VS Code terminal. Use your own email and name:

```
git config --global user.email "your@email.com"
git config --global user.name "Your Full Name"
```

### 8.2 Step 2 — Build the Final Website

This creates the `docs/` folder containing all your website files:

```
quarto render
```

**What happens:** Quarto reads all your `.qmd` files and builds the complete website into a folder called `docs/`. You will see this folder appear in your VS Code file panel on the left.

### 8.3 Step 3 — Connect Your Folder to GitHub

Run these two commands (replace `yourusername` with your actual GitHub username):

```
git init
git remote add origin https://github.com/yourusername/yourusername.github.io.git
```

### 8.4 Step 4 — Upload to GitHub

```
git add .
git commit -m "My portfolio first version"
git push -u origin main
```

**Authentication:** A browser window may open asking you to authorise Git Credential Manager. Click “**Authorise git-ecosystem**”. This is completely safe and only happens once.

## 8.5 Step 5 — Configure GitHub Pages

**Step 1** — On GitHub.com, open your repository.

**Step 2** — Click **Settings** in the top menu.

**Step 3** — In the left sidebar, click **Pages**.

**Step 4** — Under “Build and deployment”:

- Source → **Deploy from a branch**
- Branch → **main**
- Folder → **/docs**
- Click **Save**

**Step 5** — Wait 2–3 minutes, then visit:

```
https://yourusername.github.io
```

**You’re live!** Your site is now publicly accessible — free, forever, no subscription, no ads.

---

## 9 Updating Your Site

After your site is live, updating it is very simple. Every time you add or change content, run these four commands:

### 9.1 The 4-Command Update Workflow

```
quarto render
git add .
git commit -m "Describe what you changed"
git push
```

Your site will update on the internet within 1–2 minutes. Refresh your browser to see the changes.

### 9.2 Common Update Tasks

| Task                | What to do                                                         |
|---------------------|--------------------------------------------------------------------|
| Add a new blog post | Create <code>posts/name/index.qmd</code> → write → run 4 commands  |
| Update your bio     | Edit <code>index.qmd</code> → save → run 4 commands                |
| Add a new paper     | Edit <code>research.qmd</code> → add entry → save → run 4 commands |
| Change your photo   | Replace <code>images/profile.png</code> → run 4 commands           |
| Fix a typo          | Edit the file → save → run 4 commands                              |

**Memorise these 4 commands:** `quarto render` → `git add .` → `git commit -m "message"` → `git push`

These 4 commands are all you need for the rest of your career to keep your portfolio updated.

---

## 10 Adding Images and Figures

Quarto supports simple images, scientific figures with captions, interactive maps, and live charts from Python or R code.

### 10.1 Simple Photo or Image

Save your image file in the `images/` folder, then add one line to your `.qmd` file:

```
![Caption for your image](images/my-photo.jpg)
```

To control the display size:

```
![My field survey in Goa](images/field.jpg){width=80%}
```

To align to the right:

```
![Caption](images/photo.jpg){width=40% fig-align="right"}
```

### 10.2 Side-by-Side Image Comparison

Perfect for before/after maps, satellite image comparisons, or treatment effects:

```
::: {layout-ncol=2}
![Before treatment](images/before.jpg)

![After treatment](images/after.jpg)
:::
```

You can also do a 3-column layout:

```
::: {layout-ncol=3}
![2021](images/map2021.jpg)

![2022](images/map2022.jpg)

![2023](images/map2023.jpg)
:::
```

### 10.3 Scientific Figure with Caption and Cross-Reference

For publication-quality figure referencing — exactly like in a research paper:

```
 ::: {#fig-ndvi}  
 
```

NDVI map of the study area, Kharif season 2023.  
Green = healthy vegetation, red = stressed/bare areas.

```
 :::
```

As clearly shown in @fig-ndvi, the spatial distribution of  
crop stress follows the drainage pattern of the region.

**Auto-numbering:** The @fig-ndvi reference automatically creates a numbered, clickable figure link. Quarto numbers all your figures automatically — just like LaTeX in a research paper.

## 10.4 Interactive Map (Python / Folium)

For geospatial researchers, you can embed a fully interactive map that visitors can zoom and click:

```
#| label: fig-map  
#| fig-cap: "Study sites, ICAR-CCARI research area, Goa"  
  
import folium  
m = folium.Map(location=[15.49, 73.82], zoom_start=11)  
  
folium.Marker(  
    [15.49, 73.82],  
    popup="ICAR-CCARI, Goa",  
    tooltip="Click for details"  
) .add_to(m)  
  
m
```

**Requirement:** Python and folium must be installed. Install folium with: `pip install folium`

## 10.5 Charts from Python

Paste your existing Python code directly into your .qmd file. Quarto runs it and shows the chart automatically — no export needed:

```
#| label: fig-yield  
#| fig-cap: "Soybean yield trend, coastal Goa research plots"  
  
import matplotlib.pyplot as plt  
  
years = [2019, 2020, 2021, 2022, 2023]  
yield_data = [2.1, 2.4, 2.0, 2.8, 3.1]
```

```
plt.figure(figsize=(8, 4))
plt.plot(years, yield_data, marker='o', color='#1D9E75', linewidth=2)
plt.fill_between(years, yield_data, alpha=0.1, color='#1D9E75')
plt.title("Soybean yield trend, Goa")
plt.xlabel("Year")
plt.ylabel("Yield (tonnes/ha)")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

## 10.6 Embedding a QGIS Map

Export your QGIS map as an interactive HTML file using the [qgis2web](#) plugin, then embed it in your portfolio:

**Step 1** – In QGIS: Plugins → qgis2web → Export to Leaflet → Save as HTML

**Step 2** – Copy the exported folder into your project (e.g., `maps/my-map/`)

**Step 3** – Embed it in any `.qmd` file:

```
```{=html}
<iframe src="maps/my-map/index.html"
        width="100%"
        height="500px"
        frameborder="0">
</iframe>
```
```

...

---

## 11 Quick Reference Card

### 11.1 Commands You Will Use Most

| Command                              | What it does                          |
|--------------------------------------|---------------------------------------|
| <code>quarto preview</code>          | Start local preview in browser        |
| <code>quarto render</code>           | Build the complete website into docs/ |
| <code>git add .</code>               | Stage all changed files               |
| <code>git commit -m "message"</code> | Save a snapshot with a description    |
| <code>git push</code>                | Upload to GitHub (site goes live)     |
| <code>mkdir foldername</code>        | Create a new folder                   |
| <code>cd foldername</code>           | Move into a folder                    |
| <code>code .</code>                  | Open current folder in VS Code        |

## 11.2 Markdown Cheat Sheet

| What you type                | What appears       |
|------------------------------|--------------------|
| <b>**bold text**</b>         | <b>bold text</b>   |
| <i>*italic text*</i>         | <i>italic text</i> |
| <b>## Heading</b>            | A section heading  |
| - item                       | Bullet point       |
| [link text](https://url.com) | Clickable link     |
| ![alt](images/photo.jpg)     | An image           |
| `code`                       | Inline code        |

## 11.3 Useful Links

| Resource               | Link                                                                                                                        |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Quarto documentation   | <a href="https://quarto.org/docs">https://quarto.org/docs</a>                                                               |
| Your live portfolio    | <a href="https://yourusername.github.io">https://yourusername.github.io</a>                                                 |
| Your GitHub repository | <a href="https://github.com/yourusername/yourusername.github.io">https://github.com/yourusername/yourusername.github.io</a> |
| Google Scholar export  | <a href="https://scholar.google.com">https://scholar.google.com</a>                                                         |
| ORCID profile          | <a href="https://orcid.org">https://orcid.org</a>                                                                           |

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